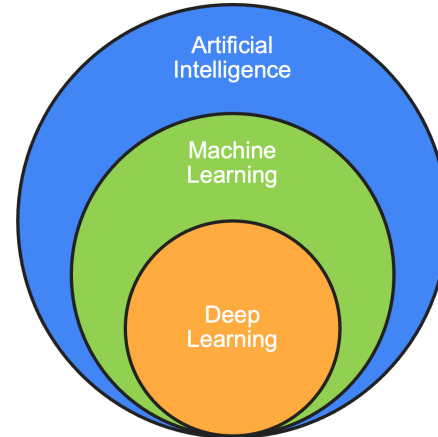
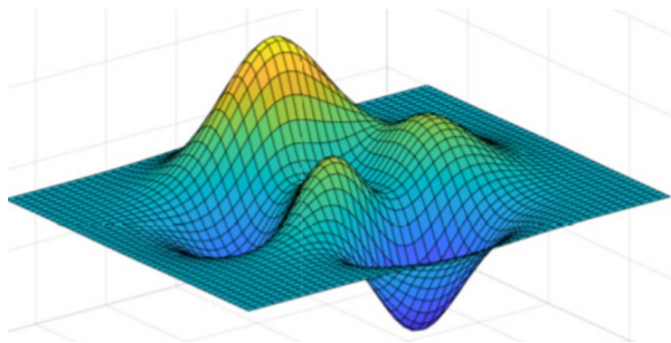


Advancements in Machine Learned Interatomic Potentials

Austin Ellis

Department of Chemistry and Biochemistry, CSUN



Outline

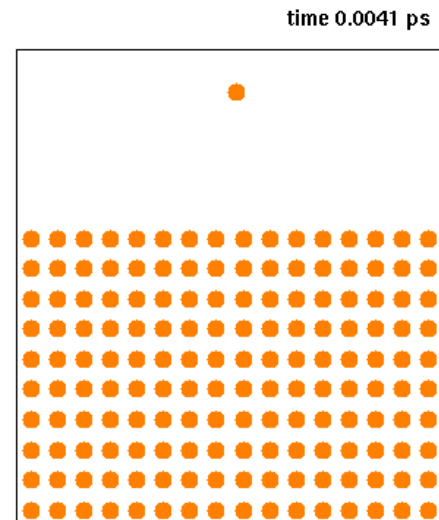
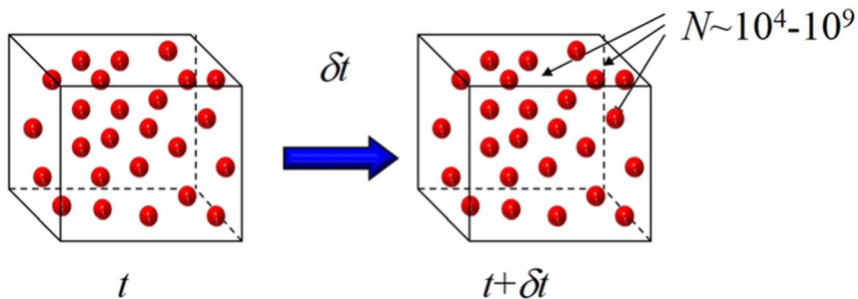
- Background
 - Interatomic Potentials: History and Trends
 - Machine Learning: Traditional vs. Deep Learning
- Machine Learned Interatomic Potentials
 - ACE: A Supervised Learning Approach
 - **MACE**: A Deep Learning Modification
- MACE Applications in Chemistry
 - 2D materials edges and defects – a story of success
 - Hydrogen under pressure – a story of failure
- Summary and outlook

Quantum chemistry: limit and beyond

Main issue: computational costs too high!

- Space scale – 1K to 1M atoms
- Time scale – picosecond to capture reactions

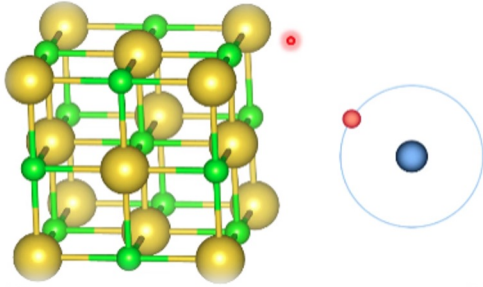
Scaling with electron count (N) means
only small systems are possible



Interatomic potentials let us model the way a system
changes over time with only energies and forces!

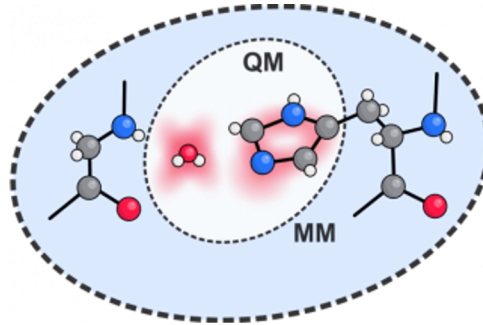
Types of simulations

First Principles (QM)



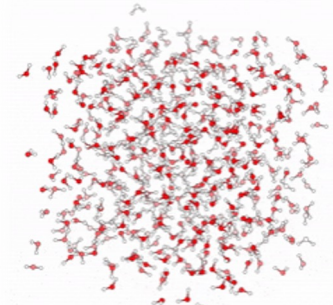
$$-\frac{\hbar^2}{2m}\nabla^2\Psi + V\Psi = E\Psi$$

QM/MM



Hybrid

Molecular Mechanics (MM)



$$F(X) = -\nabla V(X) = MV(t)$$
$$V(t) = X(t)$$

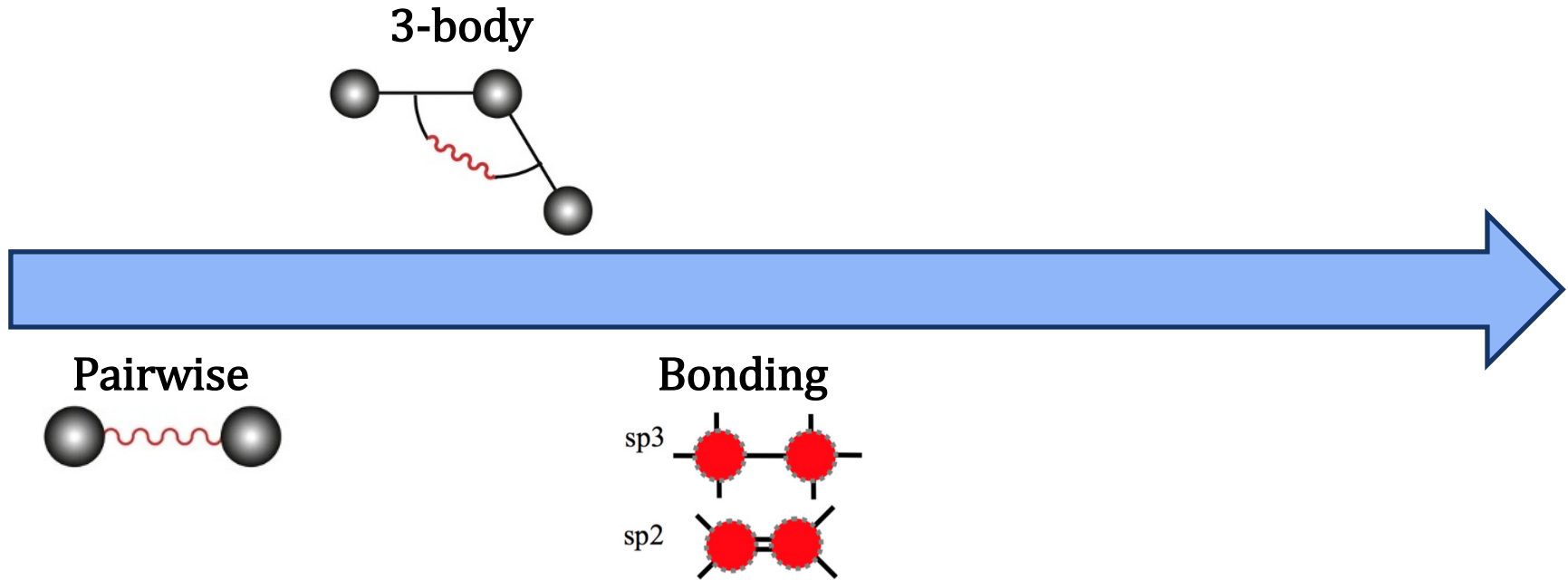
Need accurate
Interatomic Potentials

Accuracy



Scalability

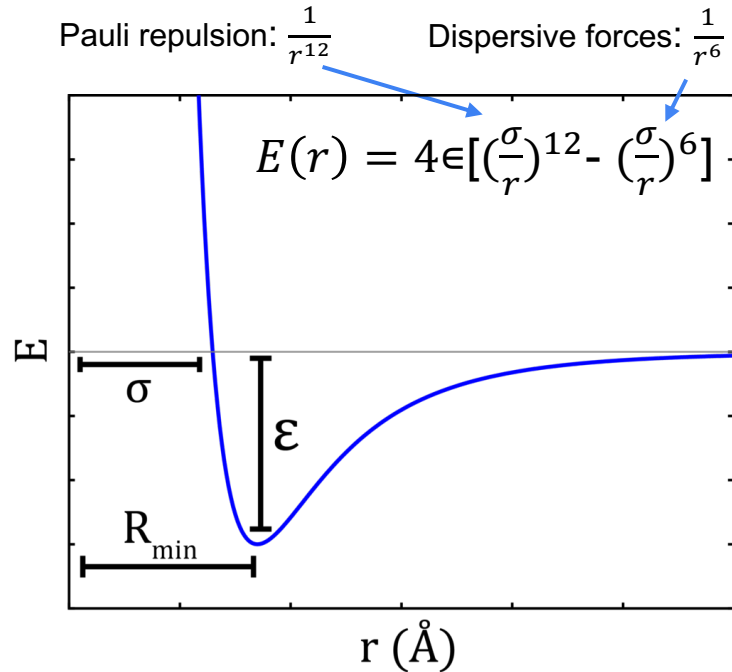
Developments of Interatomic potentials



A major challenge of Interatomic Potentials is to capture **Chemistry**

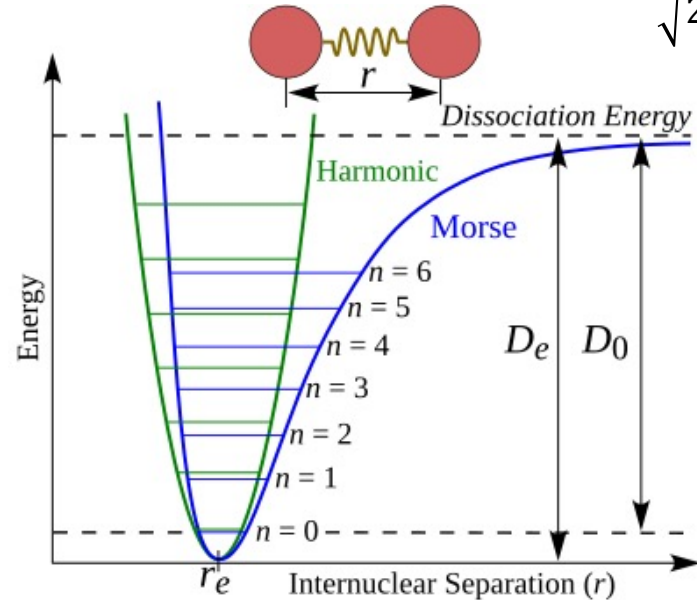
Semi-empirical diatomic potentials

Lennard-Jones Potential (1924)



Morse Potential (1929)

$$V(r) = D_e(1 - e^{-\alpha(r-r_e)})^2 \quad \alpha = \sqrt{\frac{k_e}{2D_e}}$$



Can be found in many textbooks 6

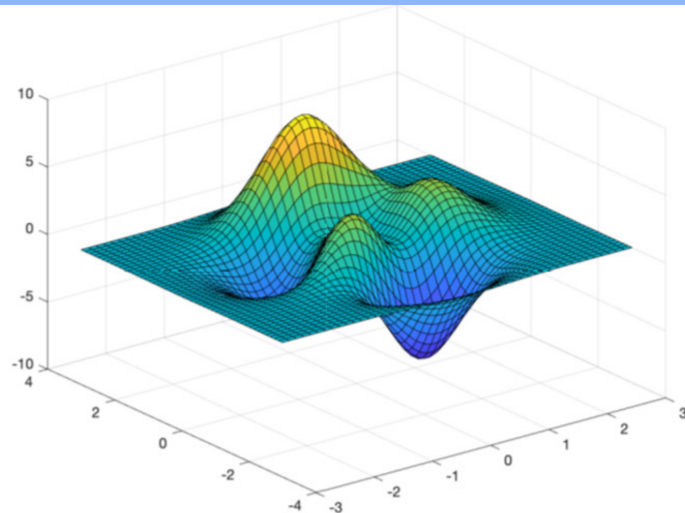
Beyond pair-wise interatomic potentials

Stillinger-Weber (1985)

$$E = \sum_i \sum_{j>i} \phi_2(r_{ij}) + \sum_i \sum_{j \neq i} \sum_{k>j} \phi_3(r_{ij}, r_{ik}, \theta_{ijk})$$

$$\phi_2(r_{ij}) = A_{ij} \epsilon_{ij} \left[B_{ij} \left(\frac{\sigma_{ij}}{r_{ij}} \right)^{p_{ij}} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^{q_{ij}} \right] \exp \left(\frac{\sigma_{ij}}{r_{ij} - a_{ij} \sigma_{ij}} \right)$$

$$\phi_3(r_{ij}, r_{ik}, \theta_{ijk}) = \lambda_{ijk} \epsilon_{ijk} [\cos \theta_{ijk} - \cos \theta_{0ijk}]^2 \exp \left(\frac{\gamma_{ij} \sigma_{ij}}{r_{ij} - a_{ij} \sigma_{ij}} \right) \exp \left(\frac{\gamma_{ik} \sigma_{ik}}{r_{ik} - a_{ik} \sigma_{ik}} \right)$$



Stillinger-Weber (SW) is a three-body potential adding angular dependence, improving pairwise models for covalent materials.

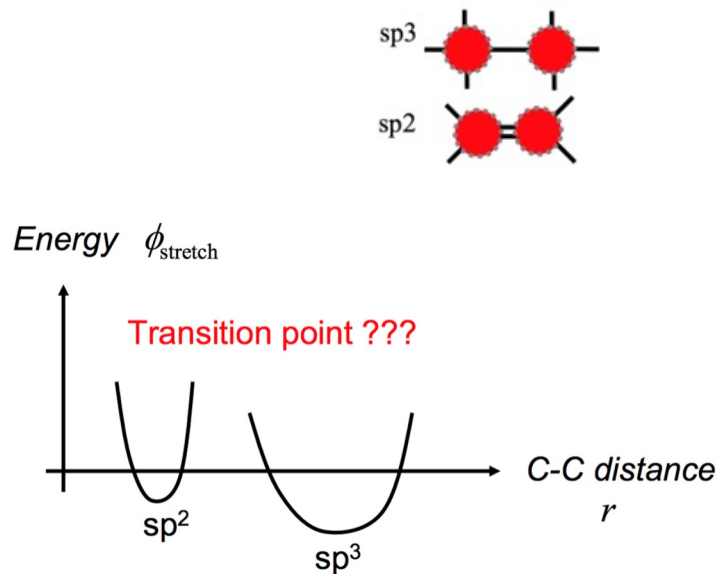
More accurate than pair potentials for materials where bond angles determine structure and properties (like the Si system was developed for)

Capturing chemistry by interatomic potentials

Reactive Force Field (ReaxFF)

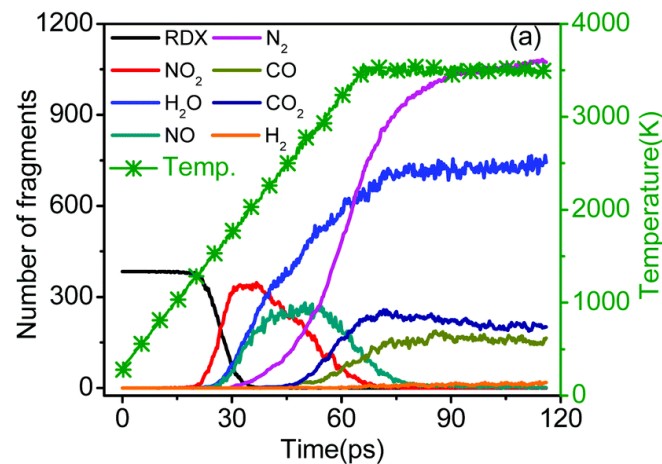
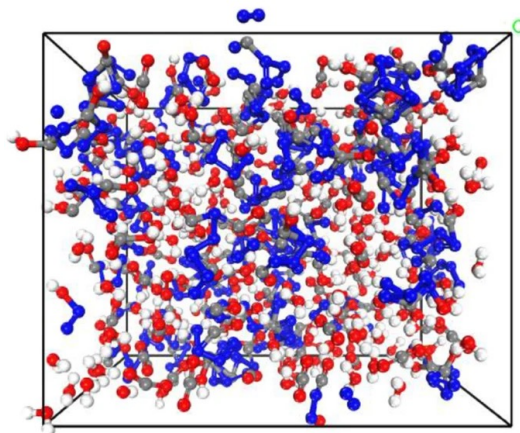
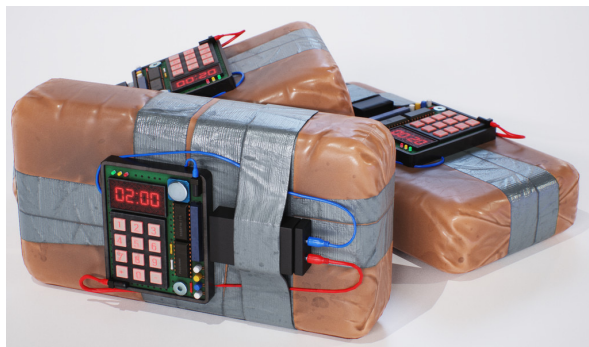
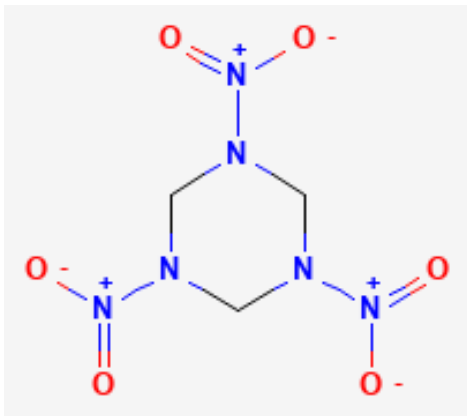
- Pairwise potentials can't predict the breaking and forming of bonds
- Bond order based force field developed at CalTech by Van Duin and Goddard (2001)

Bond length $\rightarrow$ Bond order $\rightarrow$ Energy



Capturing chemical reactions

ReaxFF simulating RDX decomposition



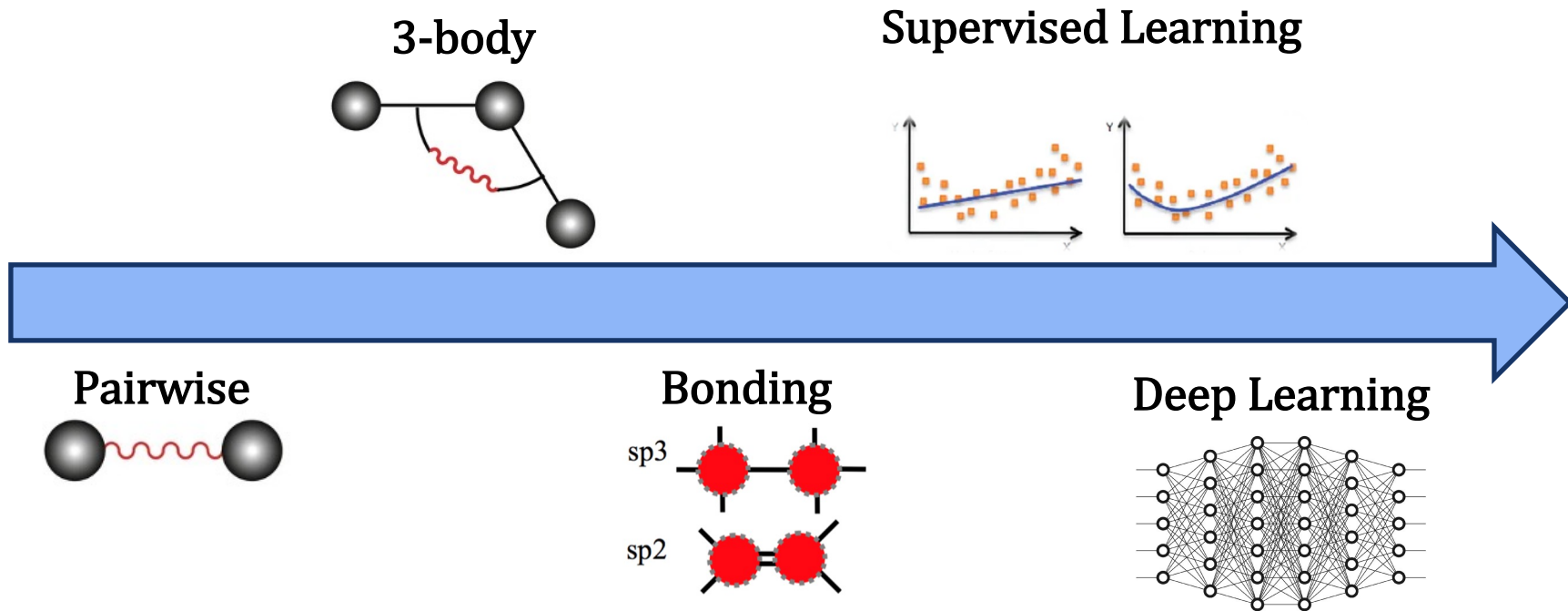
Three birds, one stone

Generally...	First Principles (DFT, etc.)	“Classical” Force Fields	Machine Learning Interatomic Potentials
Fast	✗	✓	✓ ?
Accurate	✓	✗	✓ ?
Scalable	✗	✓	✓ ?



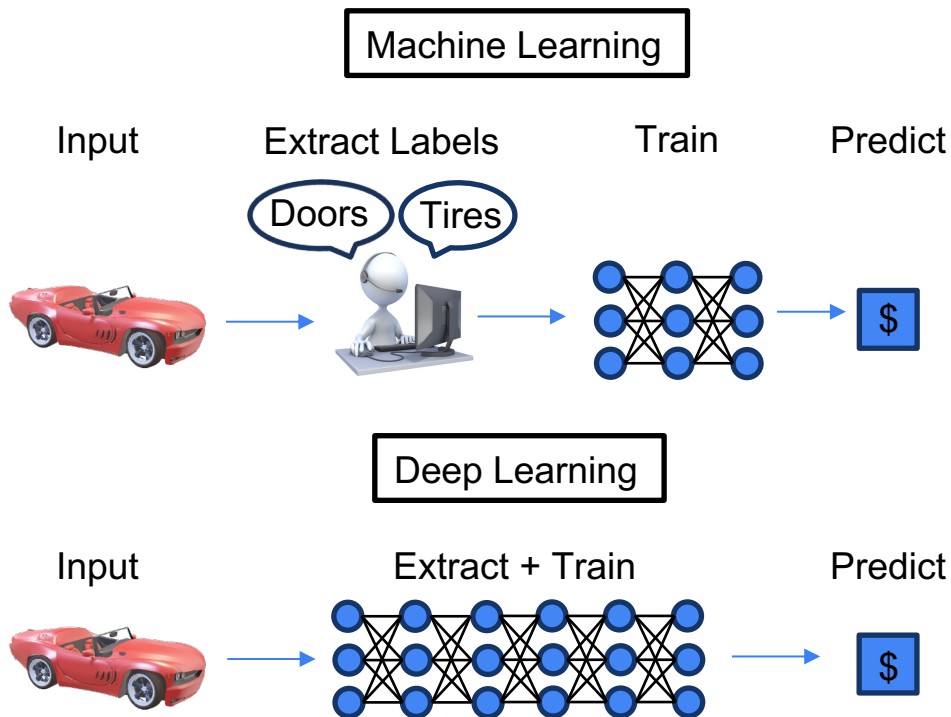
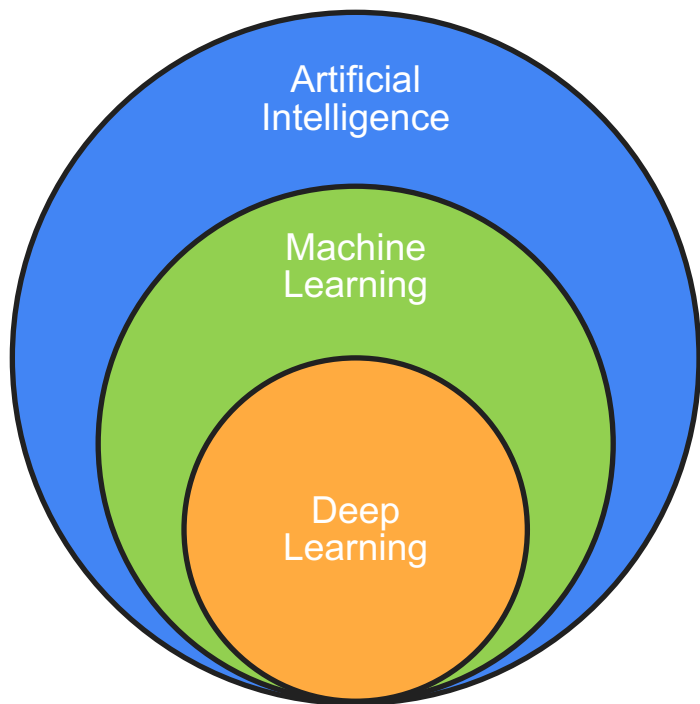
How does it work?

Paradigm shift: Data Driven Potentials

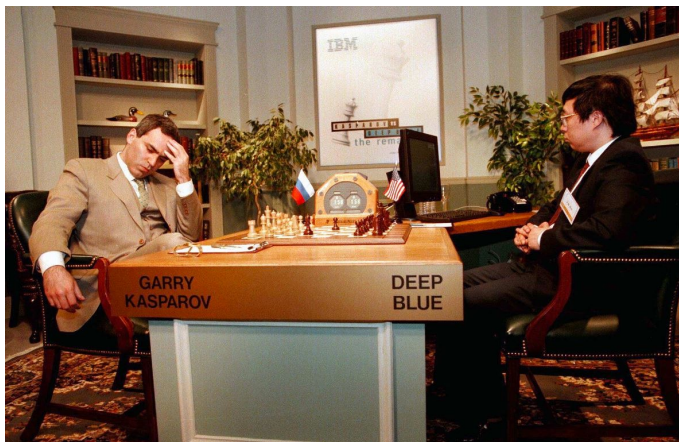


Machine learning methods

ML advancements: old versus deep

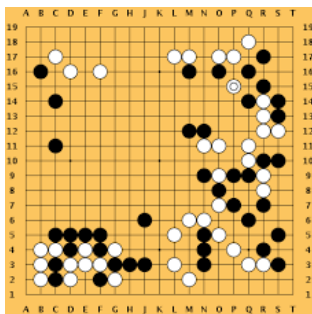


ML history: old versus deep



1996

Human
vs.
Computer



2016



ML: from chess to GO ...

Varying Complexities



Elements

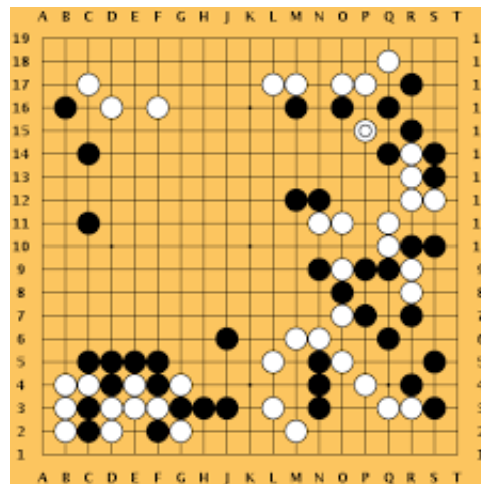
32, 6

Structures

2D, 8×8

Rules

moderate



<361, 2

2D, 19×19 2.08×10^{170}

simple

... to chemistry

How about chemistry?

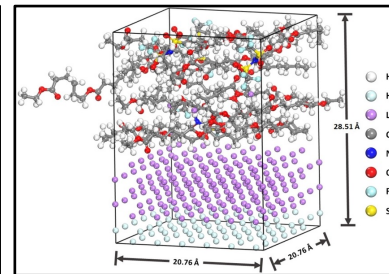
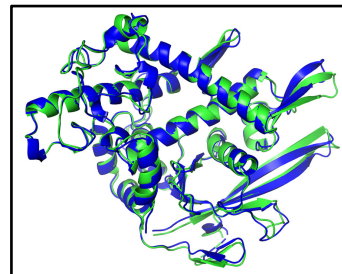
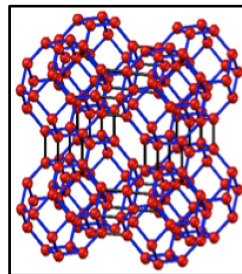
Elements

100+

1 H 1.008																				2 He 4.0026															
3 Li 6.94		4 Be 9.0122												5 B 10.81		6 C 12.011		7 N 14.007		8 O 15.999		9 F 18.998		10 Ne 20.180											
11 Na 22.990		12 Mg 24.305												13 Al 26.982		14 Si 28.085		15 P 30.974		16 S 32.06		17 Cl 35.45		18 Ar 39.948											
19 K 39.098		20 Ca 40.078		21 Sc 44.956		22 Ti 47.867		23 V 50.942		24 Cr 51.996		25 Mn 54.938		26 Fe 55.845		27 Co 58.933		28 Ni 58.693		29 Cu 63.546		30 Zn 65.38		31 Ga 69.723		32 Ge 72.63		33 As 74.922		34 Se 78.96		35 Br 79.904		36 Kr 83.798	
37 Rb 85.468		38 Sr 87.62		39 Y 88.906		40 Zr 91.224		41 Nb 92.906		42 Mo 95.96		43 Tc [97.91]		44 Ru 101.07		45 Rh 102.91		46 Pd 106.42		47 Ag 107.87		48 Cd 112.41		49 In 114.82		50 Sn 118.71		51 Sb 121.76		52 Te 127.60		53 I 126.90		54 Xe 131.29	
55 Cs 132.91		56 Ba 137.33		71 Lu 174.97		72 Hf 178.49		73 Ta 180.95		74 W 183.84		75 Re 186.21		76 Os 190.23		77 Ir 192.22		78 Pt 195.08		79 Au 196.97		80 Hg 200.59		81 Tl 204.38		82 Pb 207.2		83 Bi 208.98		84 Po [209]		85 At [209.98]		86 Rn [222.02]	
87 Fr [223.02]		88 Ra [226.03]		103 Lr [262.11]		104 Rf [265.12]		105 Db [268.13]		106 Sg [271.13]		107 Bh [270]		108 Hs [277.15]		109 Mt [276.15]		110 Ds [281.16]		111 Rg [280.16]		112 Cn [285.17]		113 Uut [284.18]		114 Fl [289.19]		115 Uup [288.19]		116 Lv [293]		117 Uus [294]		118 Uuo [294]	

Structures

3D, infinite
configuration
space



Rules

Quantum mechanics (subtle and hard to learn)



(ACE)

Atomic Cluster Expansion: A Traditional ML Approach

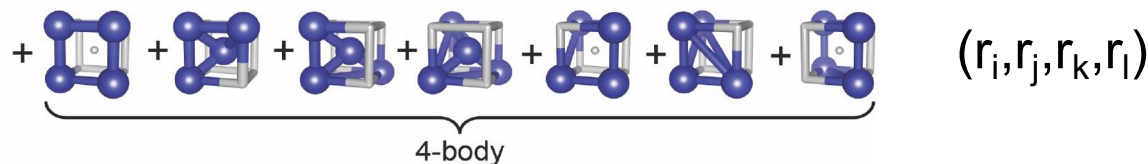
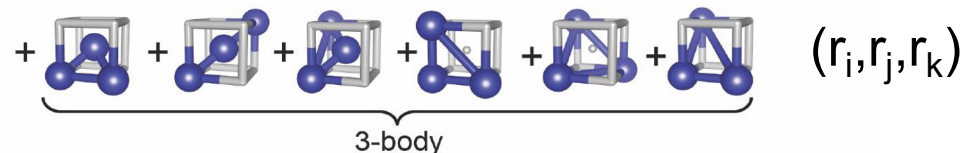
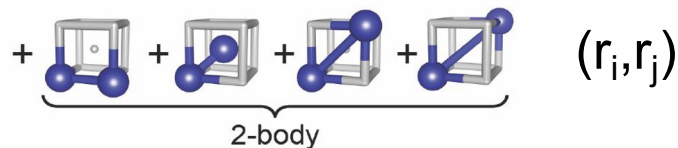
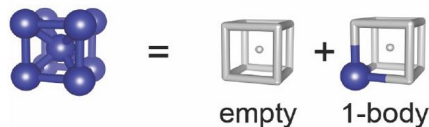
Kovacs et al. J. Chem. Theory Comput. (2021)

ACE representation

Cluster Expansion:

$$E_{\text{total}} = \sum_i V^{(1)}(\mathbf{r}_i) + \sum_{ij} V^{(2)}(\mathbf{r}_i, \mathbf{r}_j) + \sum_{ijk} V^{(3)}(\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k) + \dots$$

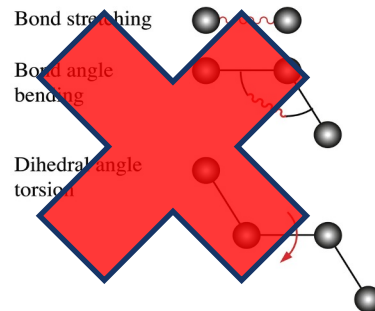
Example: BCC Lattice



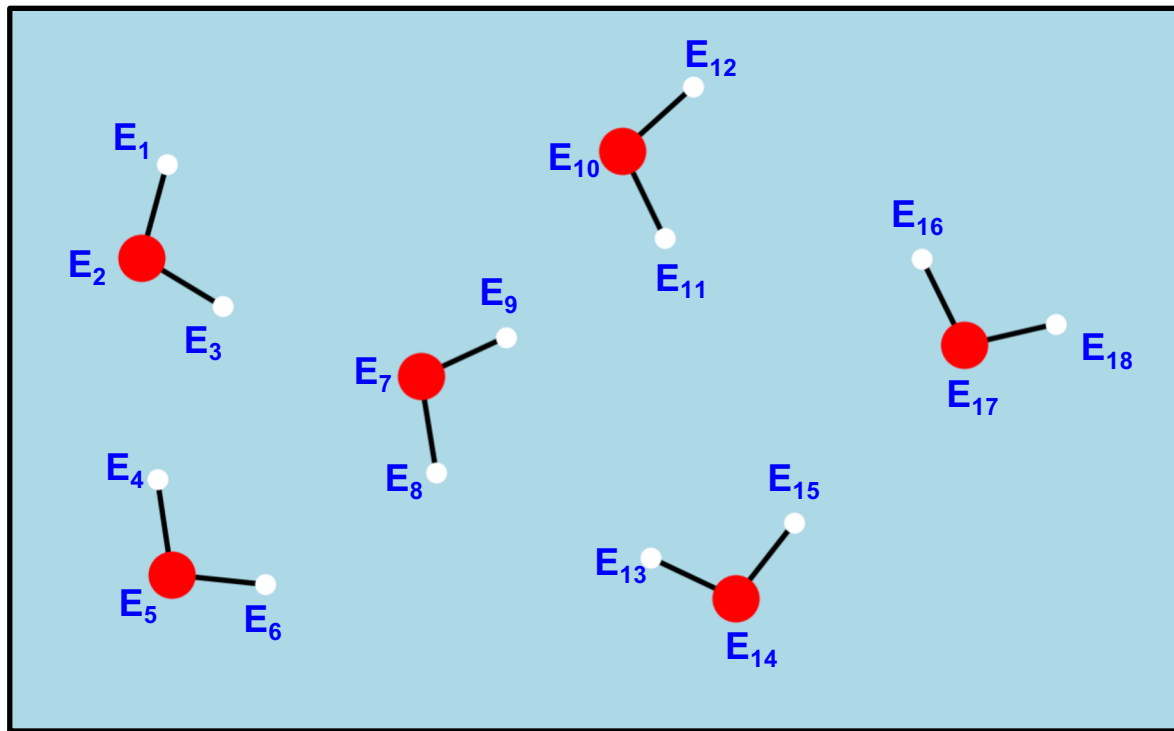
+ ...

$$E_i = V^{(1)}(\mathbf{r}_i) + \frac{1}{2} \sum_j V^{(2)}(\mathbf{r}_i, \mathbf{r}_j) + \frac{1}{3!} \sum_{jk} V^{(3)}(\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k) + \dots$$

$$E_{\text{total}} = \sum_{i=1}^N E_i$$



ACE for molecular systems



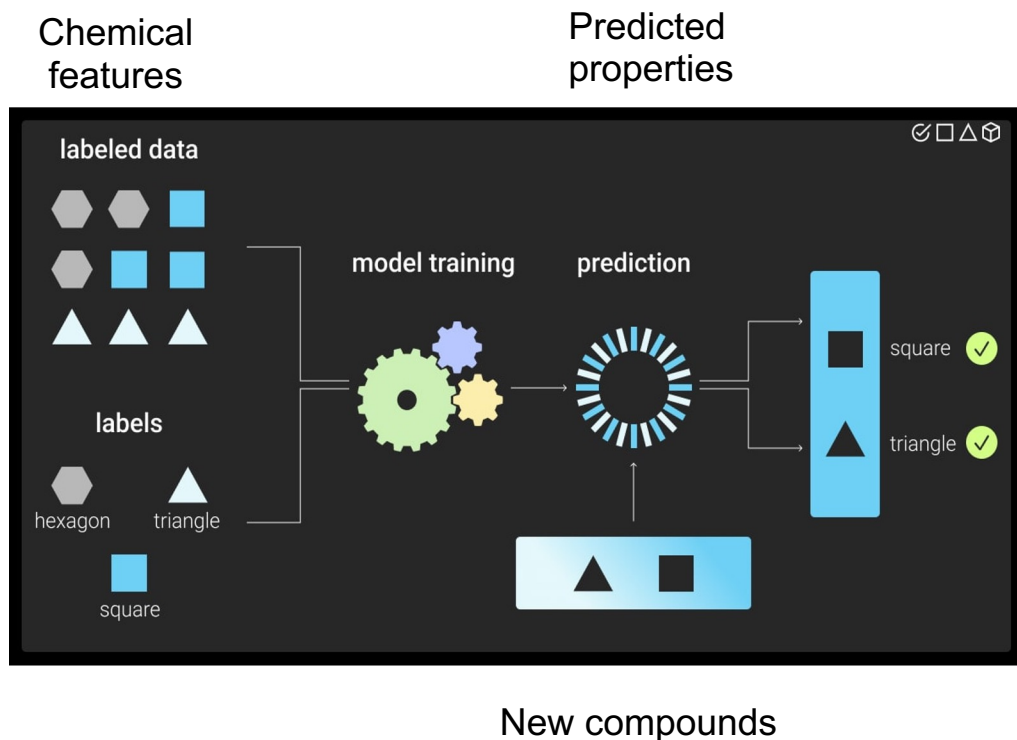
Total energy decomposes into site energies E_i

$$E_{\text{total}} = \sum_{i=1}^N E_i$$

By summing all the energy values at each site, we find the total energy of our system

What is our site energy?

“Traditional” Machine Learning

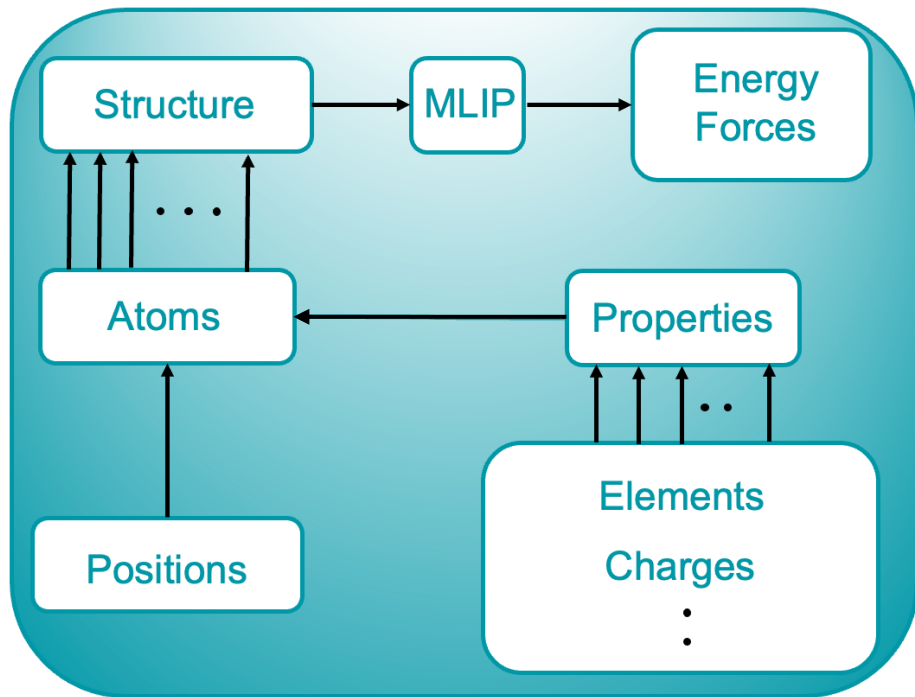


We need to think about the system we are working with and determine what terms are most relevant to its behavior

Extracted data will vary based on whether we are predicting the behavior of a:

- Protein
- 2D material
- Battery

Representing Chemical Structures



A complex example...

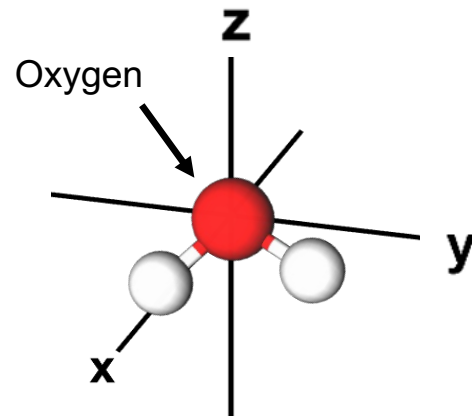
Element: 8

Charge: -2

Electronegativity: 3.5

Position: (0,0,0)

etc.



ACE Training and Predictions

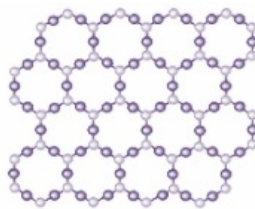
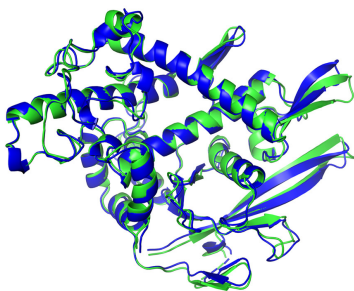
We need data:

$$\mathcal{D}_{\text{train}} = \{(R_j, \mathcal{E}_j, \mathcal{F}_j, \dots)\}$$

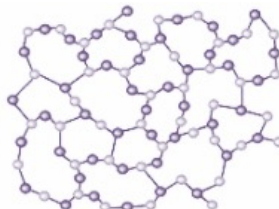
Atomic Configuration Forces

Energy Chosen features/descriptors

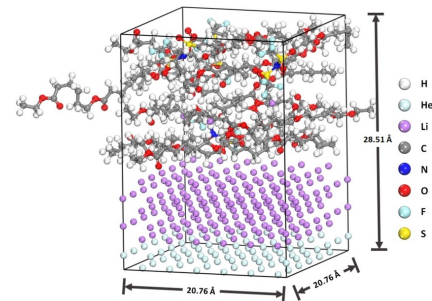
Gathered from experiment and calculation



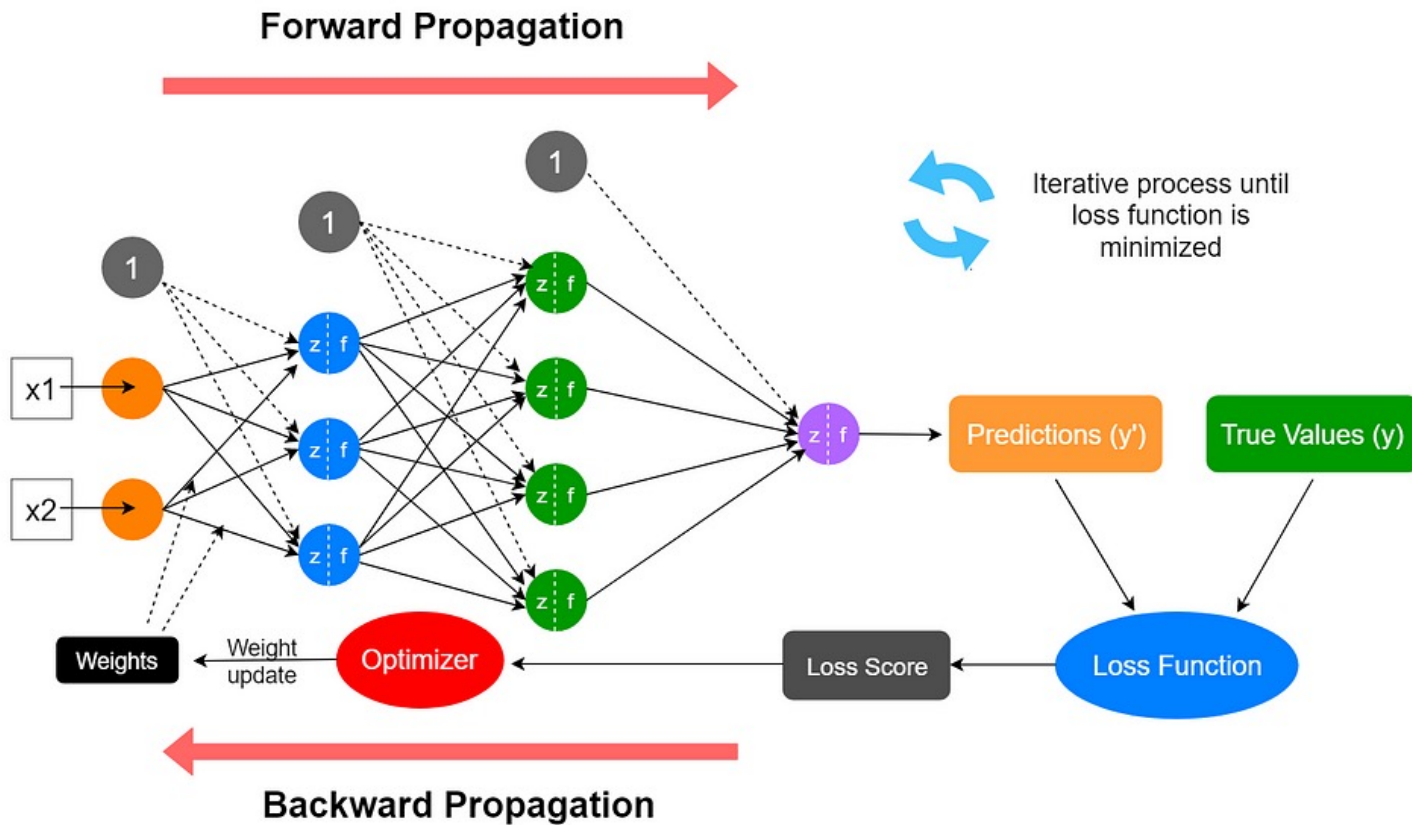
Crystalline



Amorphous

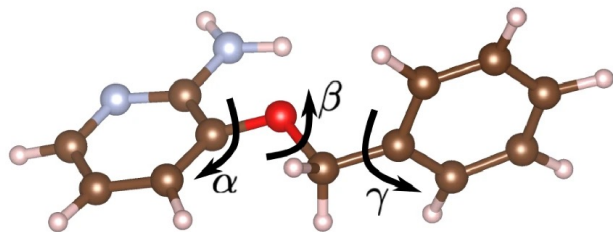


Anatomy of a Neural Network

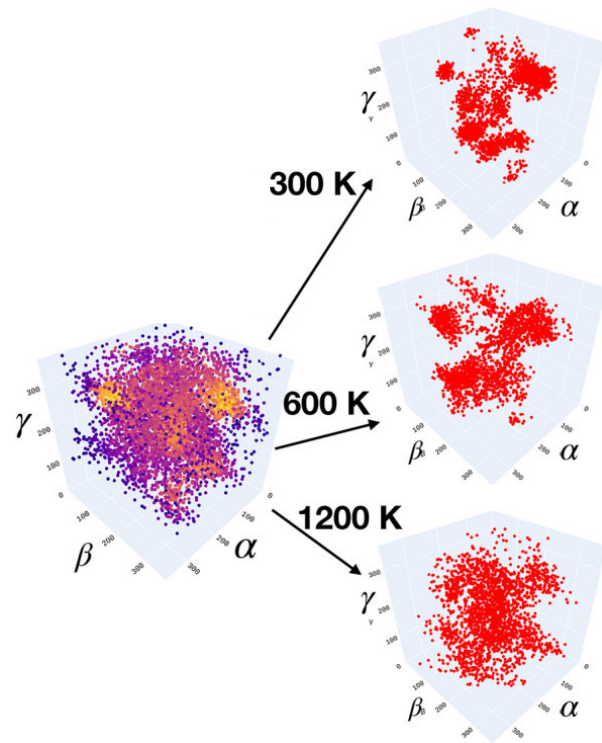
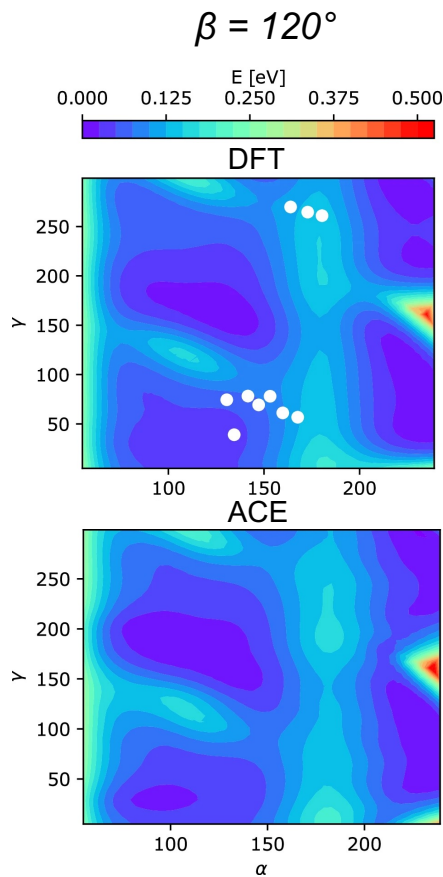


ACE applications in chemistry

Complex Dihedral Interactions



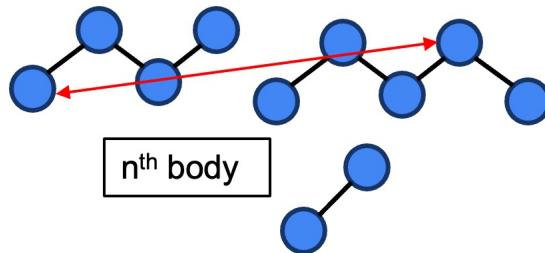
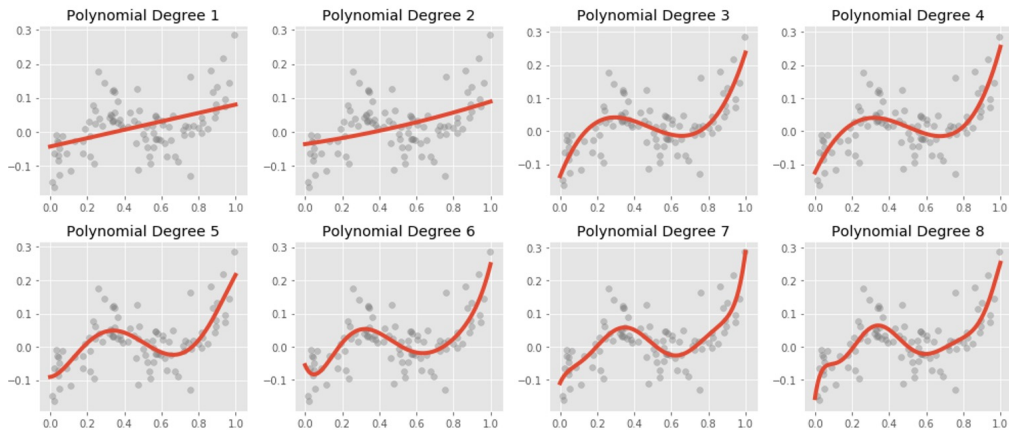
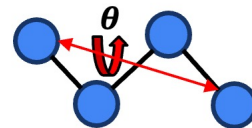
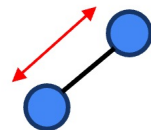
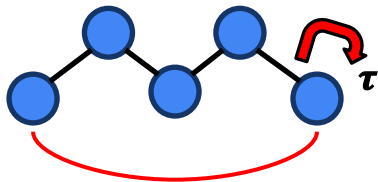
Potential Energy
Surface Prediction



Cheng npj. Comput. Mater. (2024)

How does it work?

Essentially a **polynomial-based** approach that efficiently incorporates higher-order **many-atom interactions**

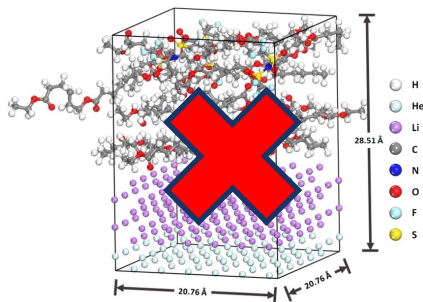
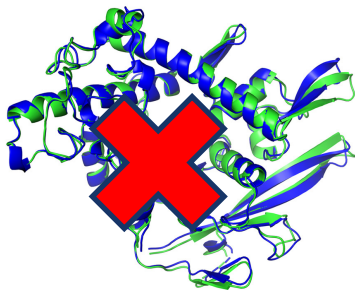


ACE Success and Limitations

ACE is a polynomial-based approach used to describe the symmetry and many-body interactions of a system that can learn to map features to properties

However....

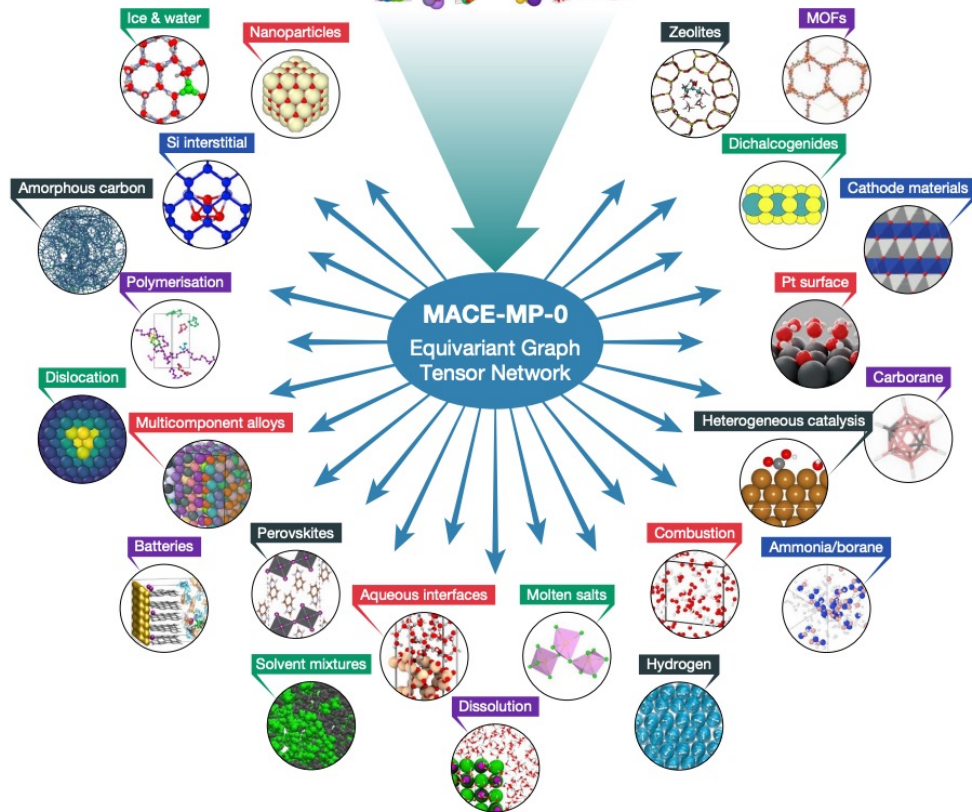
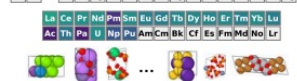
- ❖ Problems with predicting properties of overly large or particularly complex systems
- ❖ Accuracy tends to drop off after three chemical species



**How do we improve
our model?**

MATERIALS PROJECT

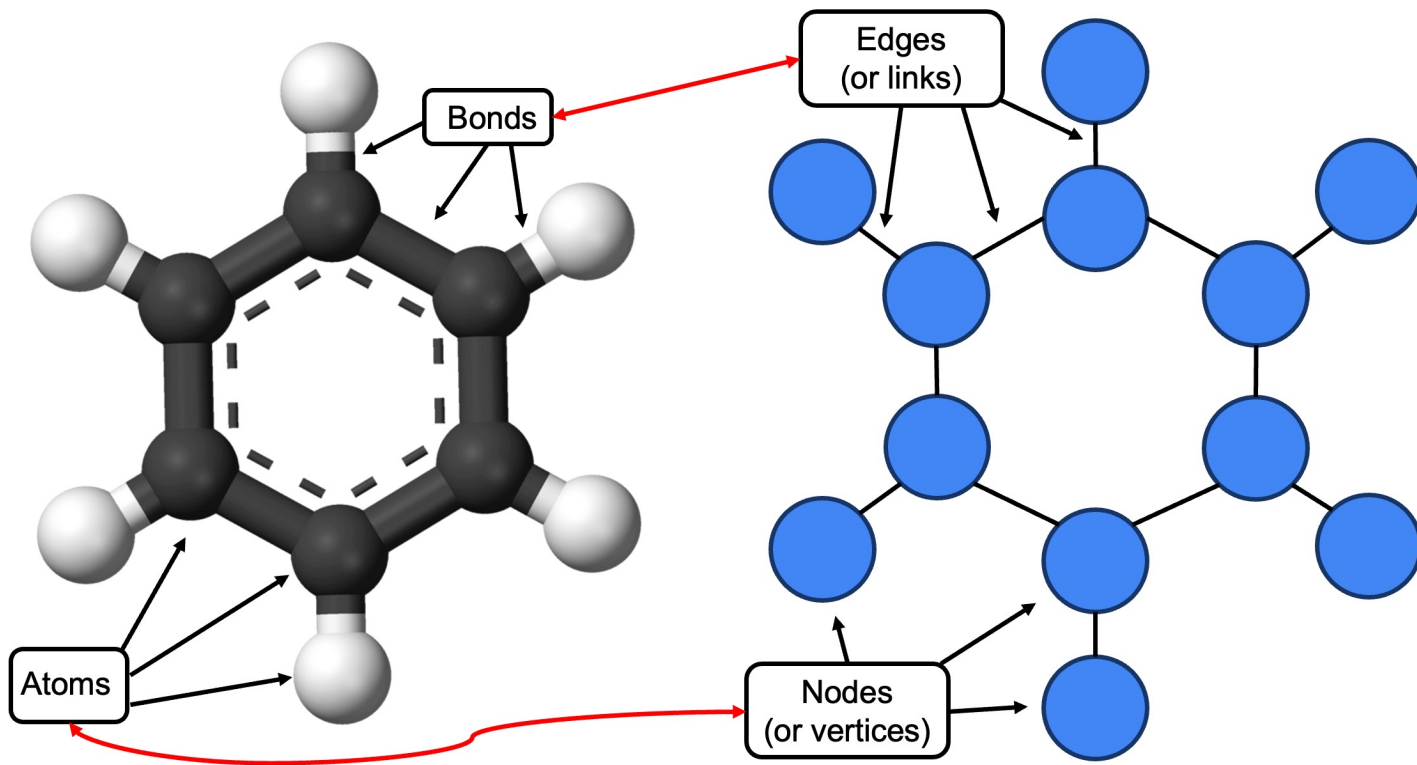
MPetri Element Counts																	
H	Li	Be	B	C	N	O	F	Ne	Na	Mg	Al	Si	P	S	Cl	Ar	Kr
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Br
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba		Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
Fr	Ra		Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Nh	Fl	Mc	Lv	Ts	Og



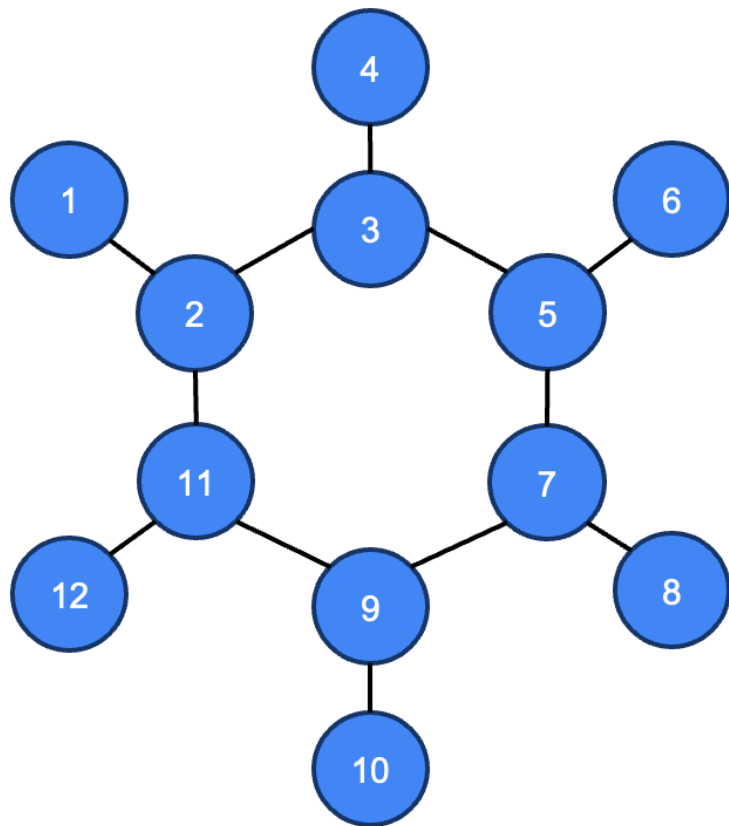
MACE

Message-passing Atomic
Cluster Expansion:
A Deep Learning Approach

From molecule to graph



Encoding atom adjacency



Adjacency Matrix

0	1	0	0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0	0	1	0
0	1	0	1	0	1	0	0	0	0	0	0
0	0	1	0	0	0	0	0	0	0	0	0
0	0	1	0	0	1	1	0	0	0	0	0
0	0	0	0	1	0	0	0	0	0	0	0
0	0	0	0	1	0	0	1	1	0	0	0
0	0	0	0	0	0	1	0	0	0	0	0
0	0	0	0	0	0	1	0	0	1	1	0
0	0	0	0	0	0	0	0	1	0	0	0
0	1	0	0	0	0	0	0	1	0	0	1
0	0	0	0	0	0	0	0	0	0	0	1

How to represent atoms

State: each node i in layer t state $\sigma_i^{(t)}$ is defined on each atom by three quantities:

$$\sigma_i^{(t)} = (r_i, z_i, h_i^{(t)})$$



$\{ x \quad y \quad z \} \quad \{ z_i \} \quad \{ h_i^{(t)} \}$



Vector

Dimension $\mathbb{R}^d$

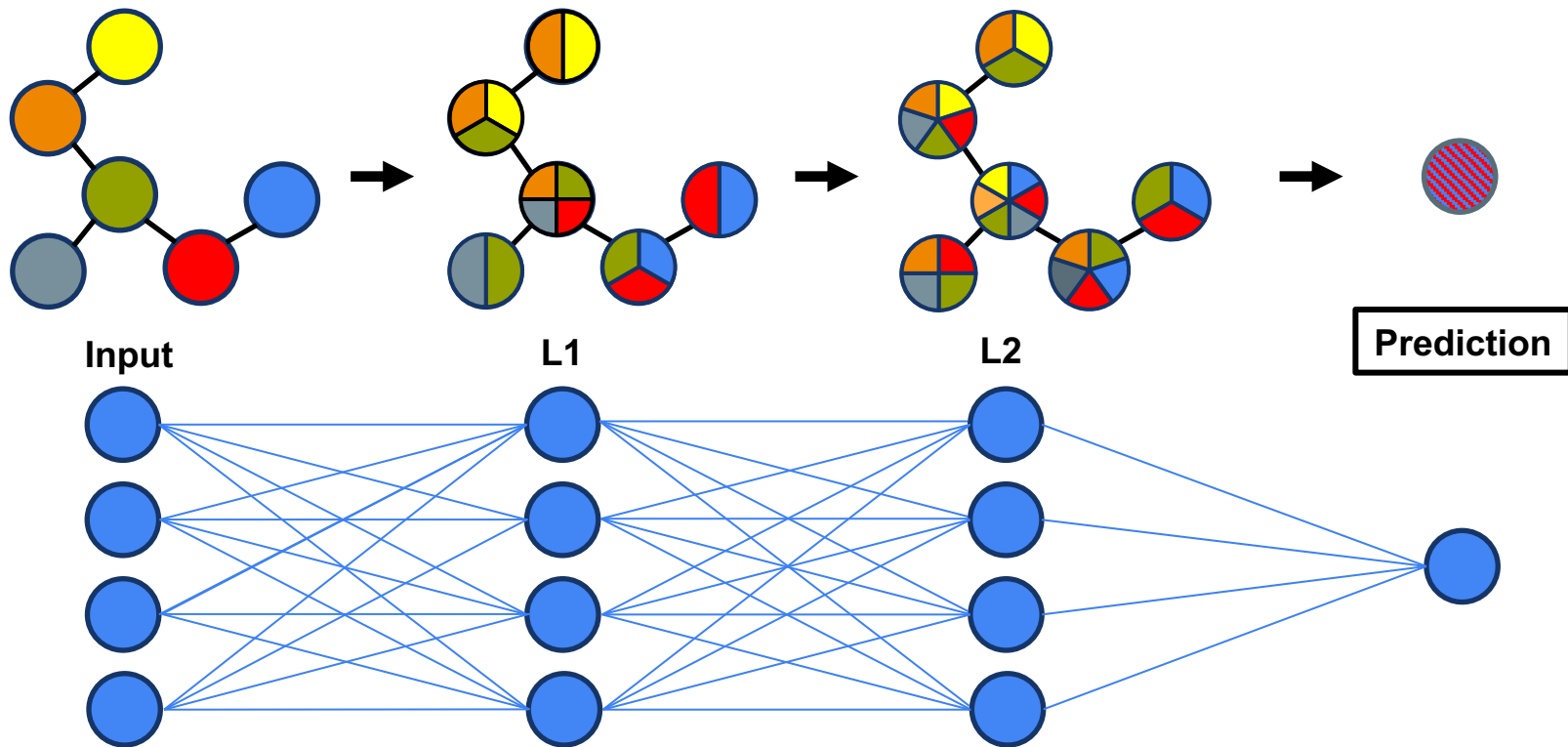
Position: r_i

Species: z_i

Learnable features: $h_i^{(t)}$

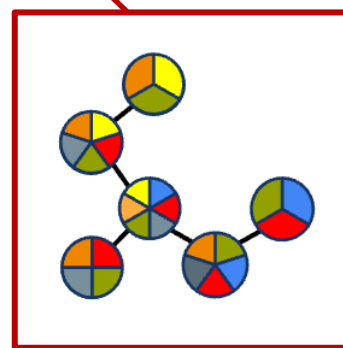
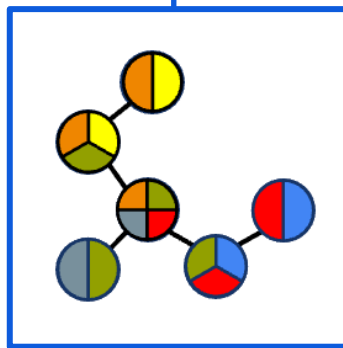
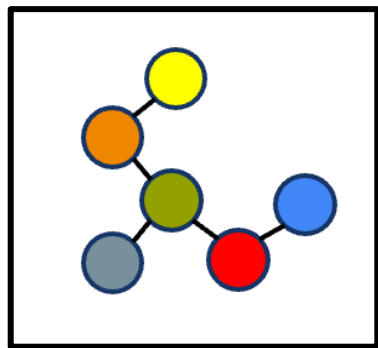
t : layer

Graph propagation across layers



Information propagation across layers

$$E_i = \boxed{V^{(1)}(\mathbf{r}_i)} + \boxed{\frac{1}{2} \sum_j V^{(2)}(\mathbf{r}_i, \mathbf{r}_j)} + \boxed{\frac{1}{3!} \sum_{j,k} V^{(3)}(\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k)} + \dots$$



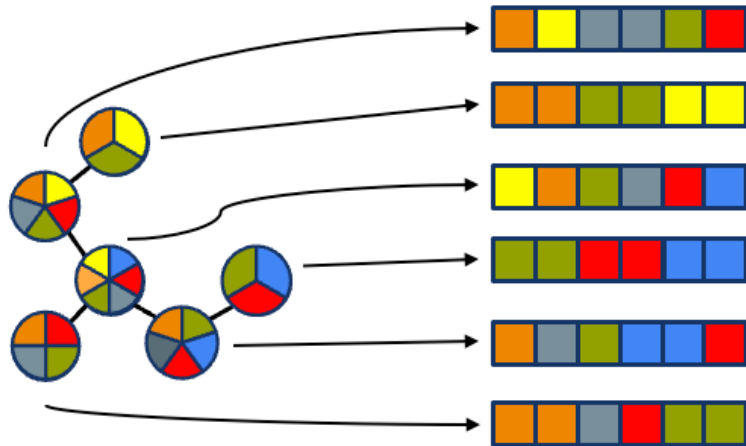
A neural network learns function u to optimally modify messages between nodes

$$m_i^{(t)} = \boxed{\sum_j u_1(\sigma_i^{(t)}; \sigma_j^{(t)})} + \boxed{\sum_{j_1, j_2} u_2(\sigma_i^{(t)}; \sigma_{j_1}^{(t)}, \sigma_{j_2}^{(t)})} + \dots + \sum_{j_1, \dots, j_\nu} u_\nu(\sigma_i^{(t)}; \sigma_{j_1}^{(t)}, \dots, \sigma_{j_\nu}^{(t)})$$

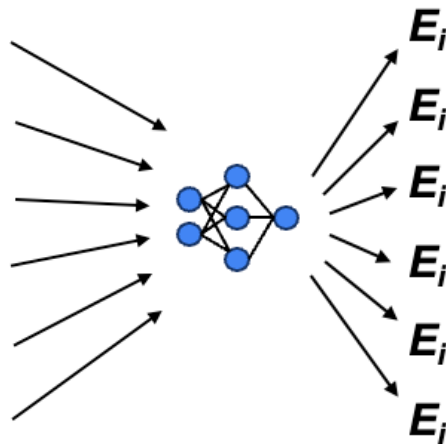


Site predictions

After final message is passed,
we can take our final node
embeddings...



Pass them into a final
neural network...

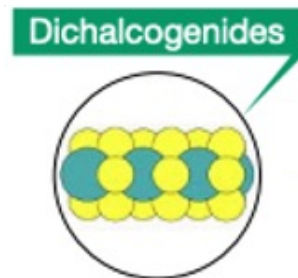


Output and sum all
the site energies!

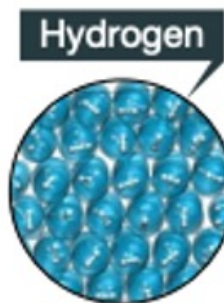
$$E_{\text{total}} = \sum_{i=1}^N E_i$$

MACE applications in chemistry

- Transition Metal Dichalcogenides
- Hydrogen under pressure
- Background
- Current Limitations
- MACE vs AIMD Analysis

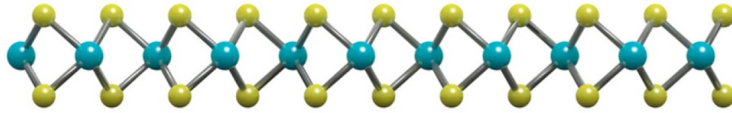


Rosen et al. J. Phys. Chem. C (2018)

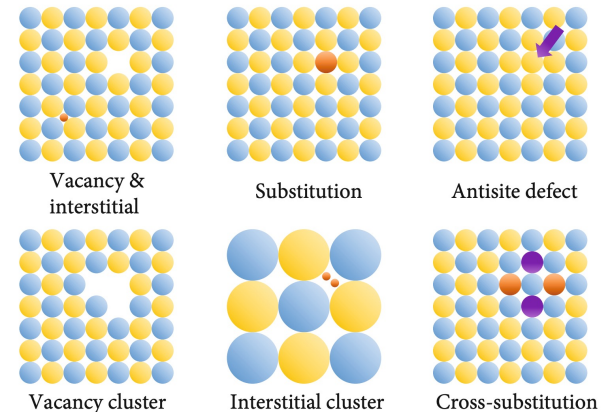
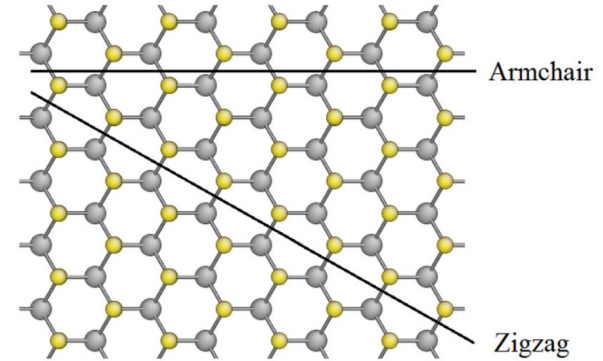


Monacelli et al. Nature (2023)

Transition metal dichalcogenides



- 2D materials made of transition metals and chalcogens
- Applied in electronics, photonics, catalysis, sensors, and energy-storage applications.
- Important structural features that modify properties
 - 1) Edges (zigzag, armchair)
 - 2) Defects (vacancies, substitutions)
- Optical properties and exciton behavior are influenced by edges and defects



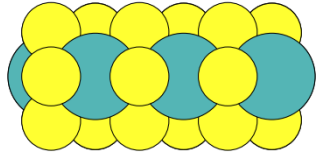
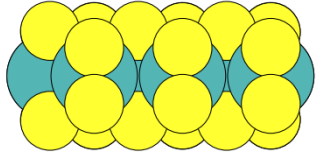
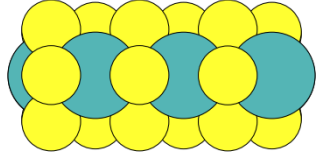
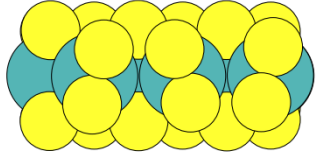
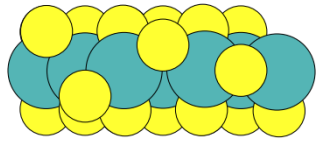
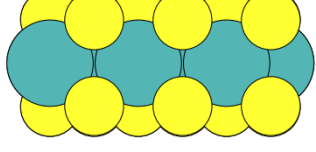
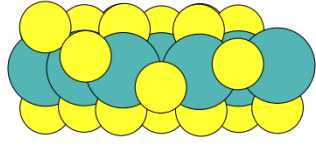
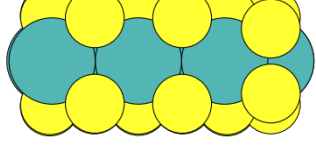
Edge structure prediction by MACE

θ_s = the sulfur coverage

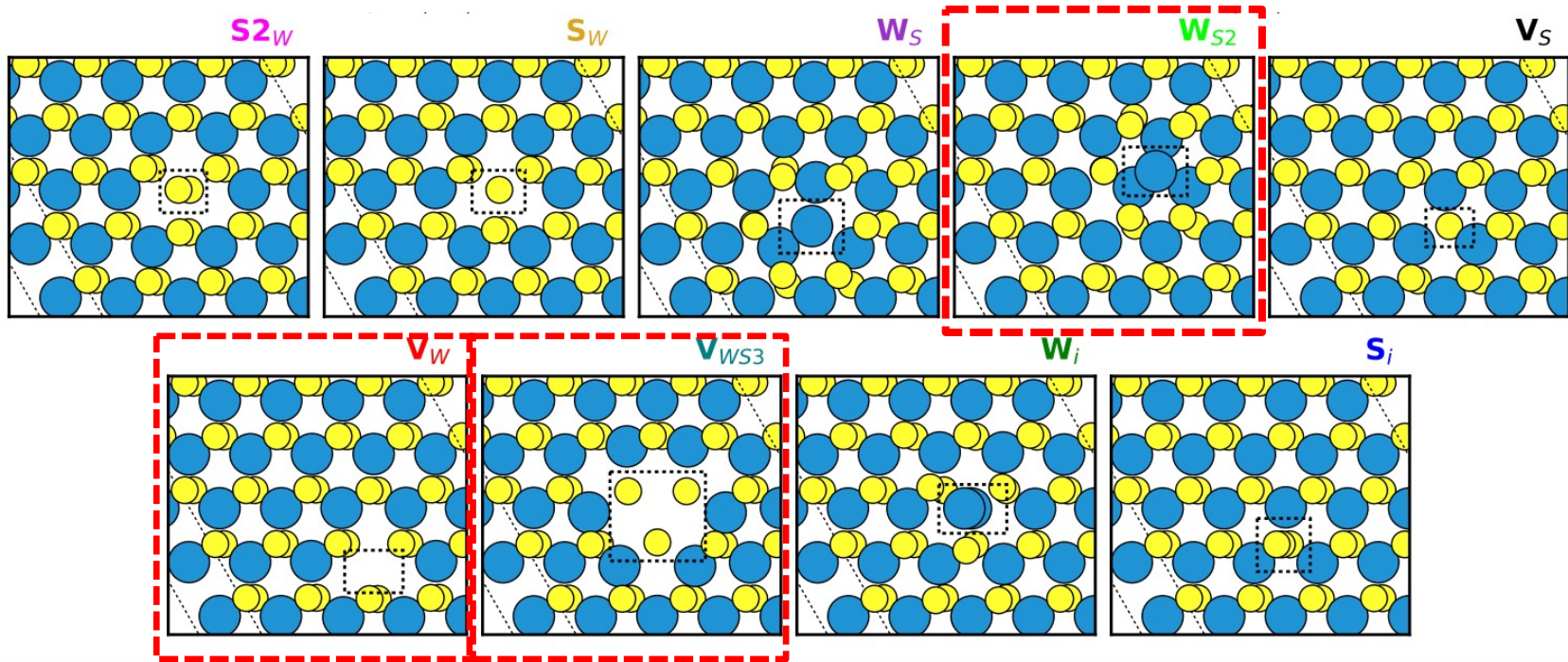
Multi-layer model with infinite stripes

Both Mo and S edges largely retain the structural features predicted by DFT

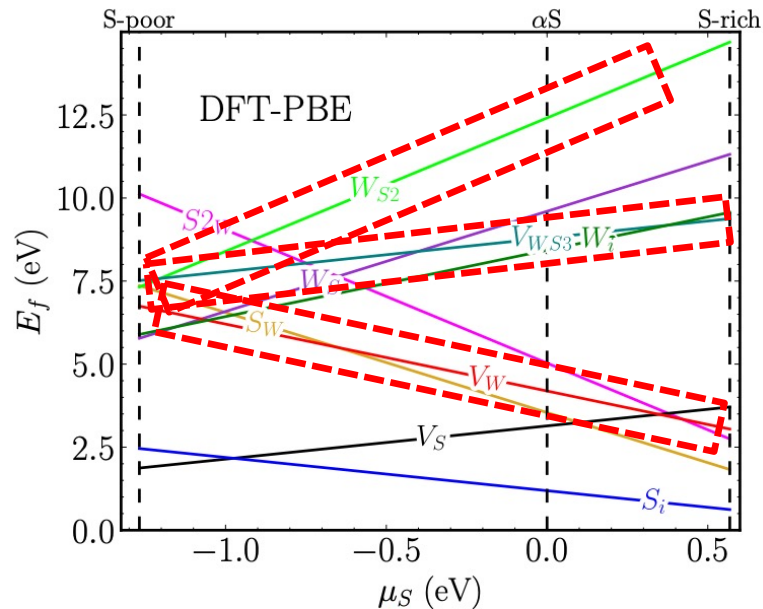
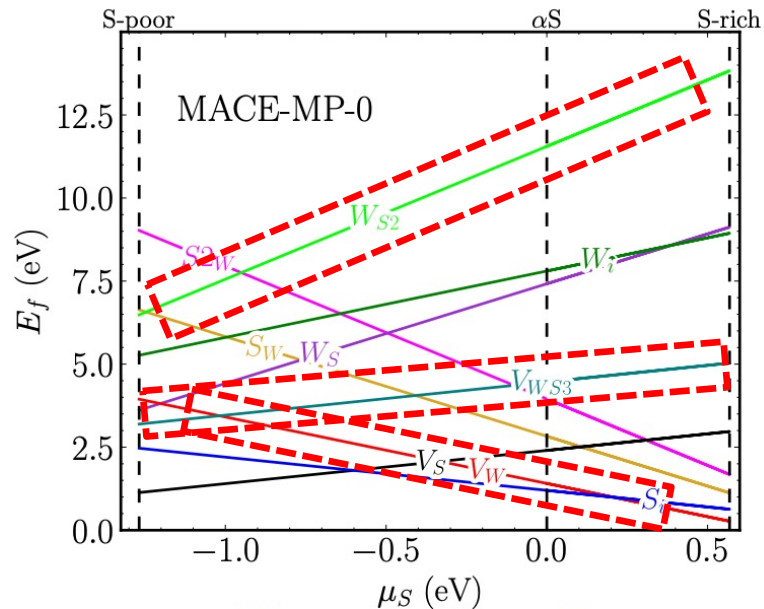
Exception of sulfur dimer formation at sulfur edge with $\theta_s=1$

		$\theta_s = 0.5$	$\theta_s = 1$
Mo-Edge	DFT-PBE		
	MACE-MP-0		
S-Edge	DFT-PBE		
	MACE-MP-0		

Defect formations in WS_2

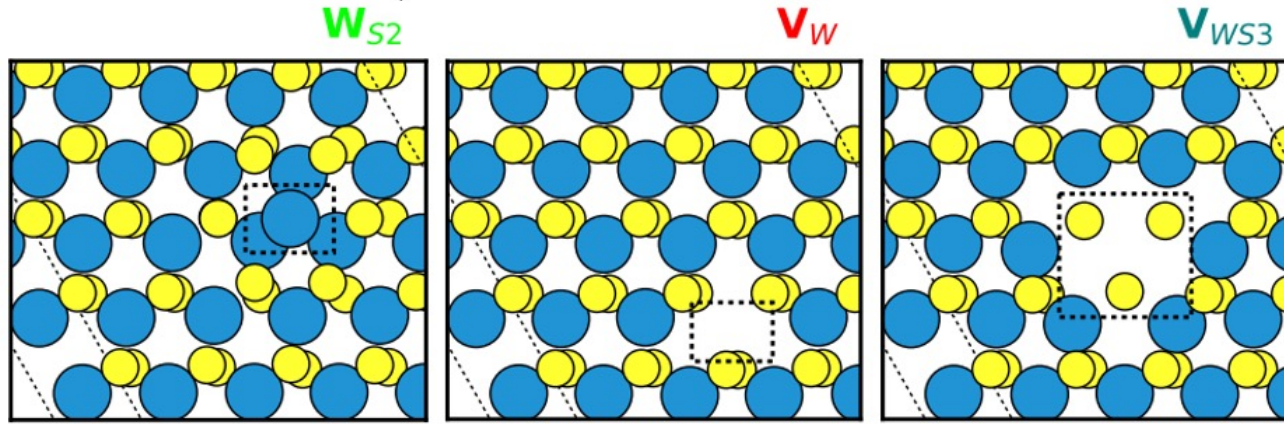


Defect formation energies



$$E_f = E_{\text{defect}} - E_{\text{pristine}} - \sum \Delta n_i \mu_i$$

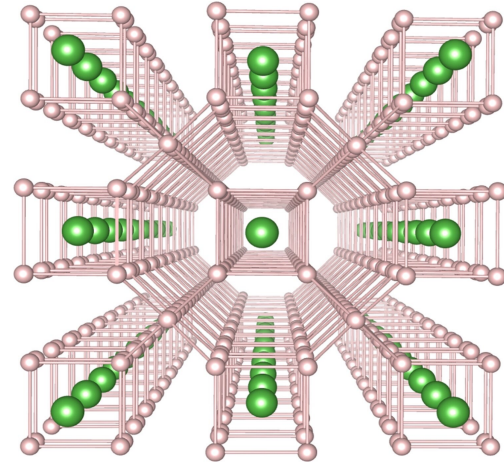
The good, bad, and ugly



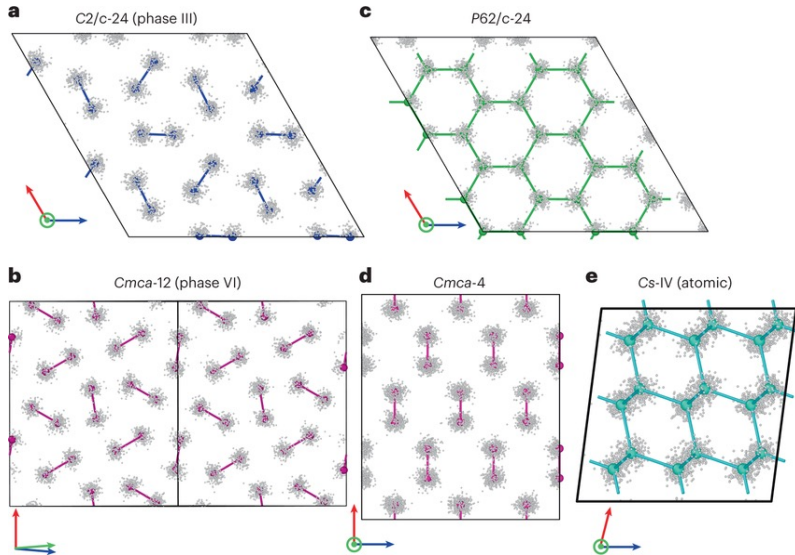
- The ability of MACE to predict defect energy formation generally related to the size of the interstitial gap left by the defect relative to ideal lattice
- Substitution of S_2 with W (large atom) yields a very accurate prediction
- Larger W and $W + S_3$ vacancies result in poor predictions due to unique chemical environment of resulting interstitial sites

Hydrogen under high pressure

- Ashcroft (1968): Predicts solid hydrogen to be a room temperature superconductor
- Success of metal superhydrides has increased interest in H_2 under pressure



Difficult to simulate due to factors like anharmonicity and zero-point energy

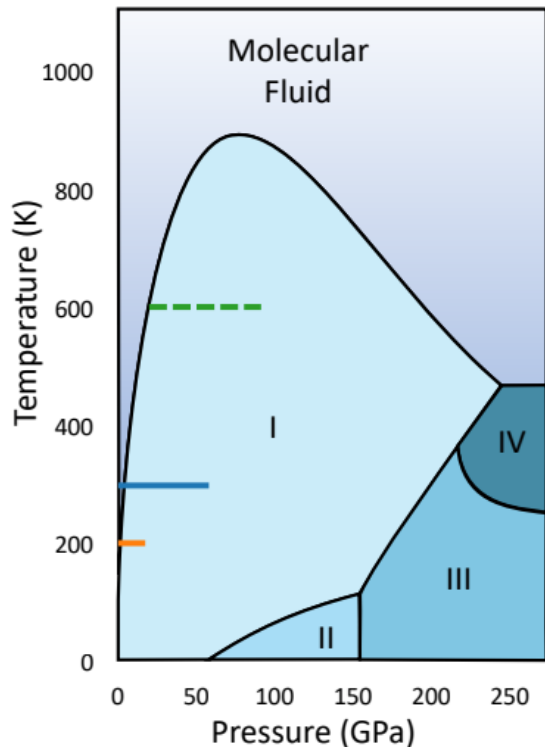


Ashcroft PRL (1968)

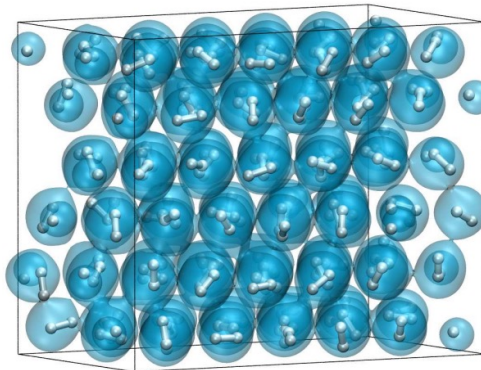
Monacelli et al. Nature (2023)

Molecular dynamics: MACE versus DFT

Illustrative phase diagram



MACE/stress
solid rotor phase I (HCP)



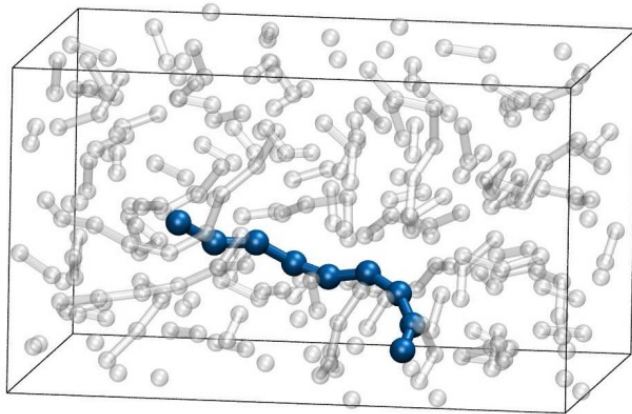
- Molecular fluid behavior (gas) behavior of hydrogen accurately predicted at low pressure
- Model predicts formation of free moving rotors in phase I
- Standard MACE model fails at 10 GPa simulation

Pickard et al. PRB (2012)

Cheng et al. Nature (2020)

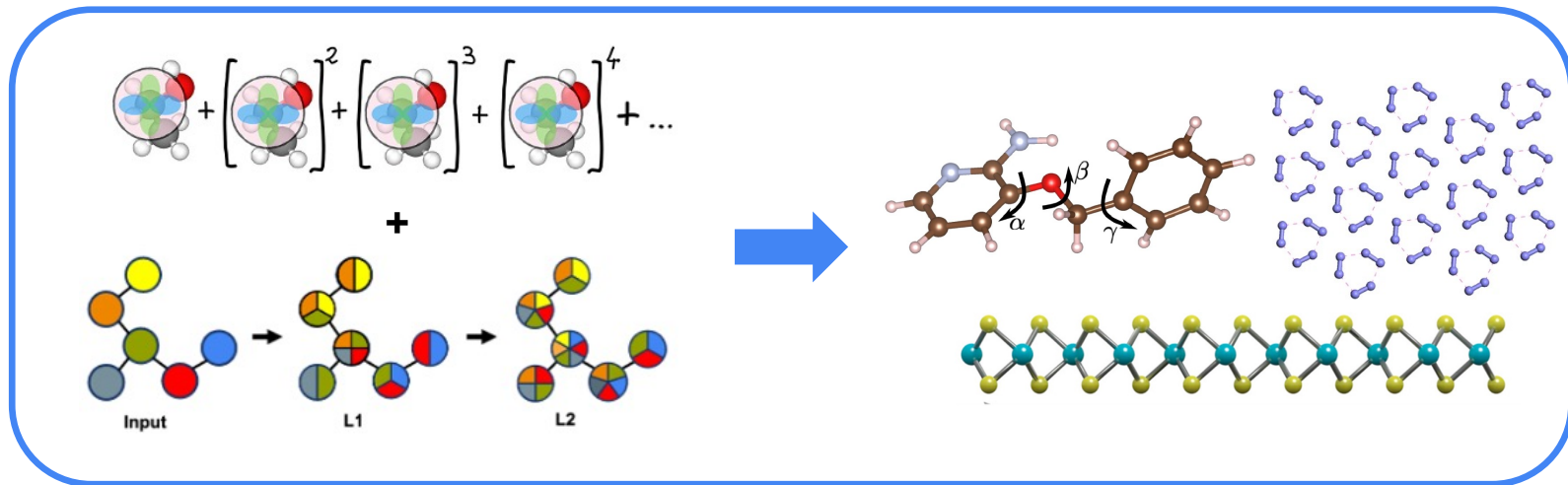
MACE high pressure failure

MACE/stress failure:
hydrogen chains (60 GPa)



- MACE-stress model created to improve prediction of hydrogen at high pressure
- Loss function modified to have a 10x multiplier on the stress error
- Improves model prediction from 10 GPa but still fails at 60 GPa via formation of non-physical hydrogen chains

Summary and Outlook



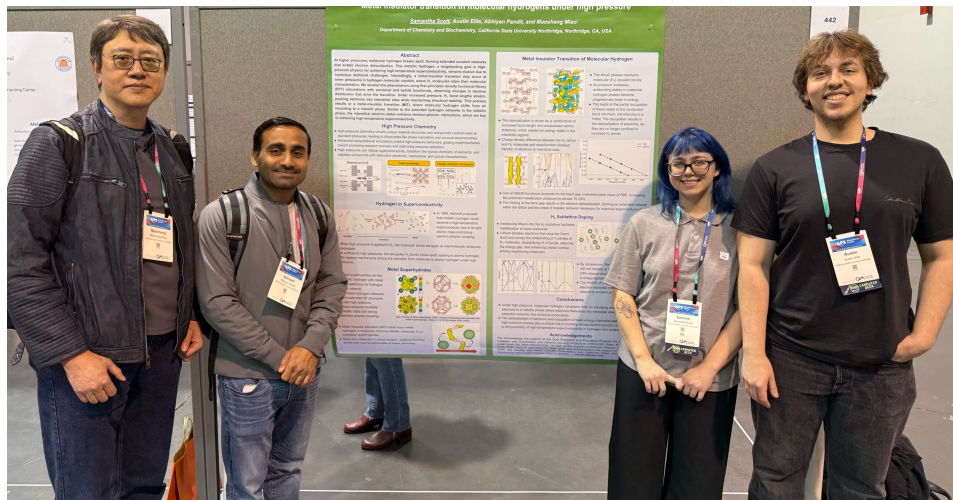
- MLIPs deliver high accuracy and computational efficiency, robustly predicting chemistry across diverse chemical and physical environments
- Accuracy dramatically goes down when applied to systems drastically different from training data

Acknowledgements

Dr. Maosheng Miao

Dr. Abhiyan Pandit

Samantha Scott



MAJOR
RESEARCH
INSTRUMENTATION



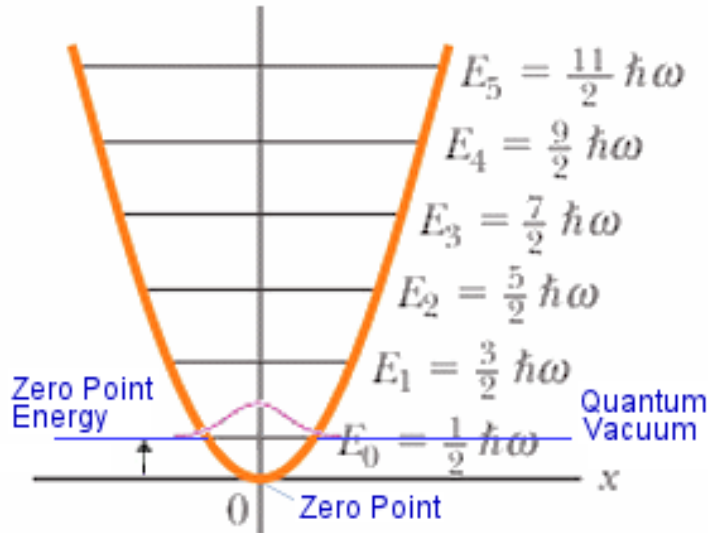
Questions?

Supplemental

Zero-point Energy

$$\frac{-\hbar^2}{2m} \frac{d^2 \Psi(x)}{dx^2} + \frac{1}{2} m \omega^2 x^2 \Psi(x) = E \Psi(x)$$

$$E_n = \hbar \omega \left(n + \frac{1}{2} \right) \quad \omega = \sqrt{\frac{k}{m}}$$



Harmonic Oscillator

Lowest possible energy a quantum system can have

Even at 0 K, systems possess residual energy due to Heisenberg uncertainty principle

MPNN Interatomic Potentials

2.1 MPNN Interatomic Potentials

$$\sigma_i^{(t)} = (\mathbf{r}_i, z_i, \mathbf{h}_i^{(t)}),$$

$$\mathbf{m}_i^{(t)} = \bigoplus_{j \in \mathcal{N}(i)} M_t(\sigma_i^{(t)}, \sigma_j^{(t)}),$$

$$\mathbf{h}_i^{(t+1)} = U_t(\sigma_i^{(t)}, \mathbf{m}_i^{(t)}),$$

$$E_i = \sum_{t=1}^T \mathcal{R}_t(\sigma_i^{(t)}).$$

MACE Message Construction

Message Construction

$$A_{i,kl_3m_3}^{(t)} = \sum_{l_1m_1, l_2m_2} C_{l_1m_1, l_2m_2}^{l_3m_3} \sum_{j \in \mathcal{N}(i)} R_{kl_1l_2l_3}^{(t)}(r_{ji}) Y_{l_1}^{m_1}(\hat{\mathbf{r}}_{ji}) \sum_{\tilde{k}} W_{k\tilde{k}l_2}^{(t)} h_{j,\tilde{k}l_2m_2}^{(t)},$$

$$A_{i,kl_1m_1}^{(1)} = \sum_{j \in \mathcal{N}(i)} R_{kl_1}^{(1)}(r_{ji}) Y_{l_1}^{m_1}(\hat{\mathbf{r}}_{ji}) W_{kz_j}^{(1)}.$$

$$\mathbf{B}_{i,\eta_\nu kLM}^{(t)} = \sum_{\mathbf{lm}} C_{\eta_\nu, \mathbf{lm}}^{LM} \prod_{\xi=1}^{\nu} \sum_{\tilde{k}} w_{k\tilde{k}l_\xi}^{(t)} A_{i,\tilde{k}l_\xi m_\xi}^{(t)}, \quad \mathbf{lm} = (l_1m_1, \dots, l_\nu m_\nu)$$

Equivariant Graph Neural Networks

2.2 Equivariant Graph Neural Networks

$$\mathbf{h}_i^{(t)}(Q \cdot (\mathbf{r}_1, \dots, \mathbf{r}_N)) = D(Q) \mathbf{h}_i^{(t)}(\mathbf{r}_1, \dots, \mathbf{r}_N),$$

$$h_{i,kLM}^{(t)}(Q \cdot (\mathbf{r}_1, \dots, \mathbf{r}_N)) = \sum_{M'} D_{M'M}^L(Q) h_{i,kLM'}^{(t)}(\mathbf{r}_1, \dots, \mathbf{r}_N),$$

MACE Update and Readout

The message $\mathbf{m}_i^{(t)}$ can now be written as a linear expansion

$$\mathbf{m}_{i,kLM}^{(t)} = \sum_{\nu} \sum_{\eta_{\nu}} W_{z_i k L, \eta_{\nu}}^{(t)} \mathbf{B}_{i, \eta_{\nu} k L M}^{(t)},$$

Update In MACE, the update is a linear function of the message and the residual connection [25]:

$$h_{i,kLM}^{(t+1)} = U_t^{kL}(\sigma_i^{(t)}, \mathbf{m}_i^{(t)}) = \sum_{\tilde{k}} W_{kL, \tilde{k}}^{(t)} m_{i, \tilde{k} L M}^{(t)} + \sum_{\tilde{k}} W_{z_i k L, \tilde{k}}^{(t)} h_{i, \tilde{k} L M}^{(t)}.$$

Readout In the readout phase, the invariant part of the node features is mapped to a hierarchical decomposition of site energies via readout functions:

$$E_i = E_i^{(0)} + E_i^{(1)} + \dots + E_i^{(T)}, \quad \text{where}$$

$$E_i^{(t)} = \mathcal{R}_t(\mathbf{h}_i^{(t)}) = \begin{cases} \sum_{\tilde{k}} W_{\text{readout}, \tilde{k}}^{(t)} h_{i, \tilde{k} 00}^{(t)} & \text{if } t < T \\ \text{MLP}_{\text{readout}}^{(t)}\left(\left\{h_{i, k 00}^{(t)}\right\}_k\right) & \text{if } t = T \end{cases}$$

Anharmonicity

The real potential energy can be expanded in the Taylor series.

$$V(R) = V(R_e) + \frac{1}{2!} \left(\frac{d^2V}{dR^2} \right)_{R=R_e} (R - R_e)^2 + \frac{1}{3!} \left(\frac{d^3V}{dR^3} \right)_{R=R_e} (R - R_e)^3 + \frac{1}{4!} \left(\frac{d^4V}{dR^4} \right)_{R=R_e} (R - R_e)^4 + \dots$$

This expansion was discussed in detail previously. The first term in the expansion is ignored since the derivative of the potential at R_e is zero (i.e., at the bottom of the well). The Harmonic Oscillator approximation only uses the next term, the quadratic term, in the series

$$V_{HO}(R) \approx V(R_e) + \frac{1}{2!} \left(\frac{d^2V}{dR^2} \right)_{R=R_e} (R - R_e)^2$$

or in terms of a spring constant (and ignore the absolute energy term) and defining r to equal the displacement from equilibrium ($r = R - R_e$), then we get the "standard" harmonic oscillator potential:

$$V_{HO}(R) = \frac{1}{2}kr^2$$

Alternatively, the expansion in Equation 5.2.5.1 can be shortened to the cubic term

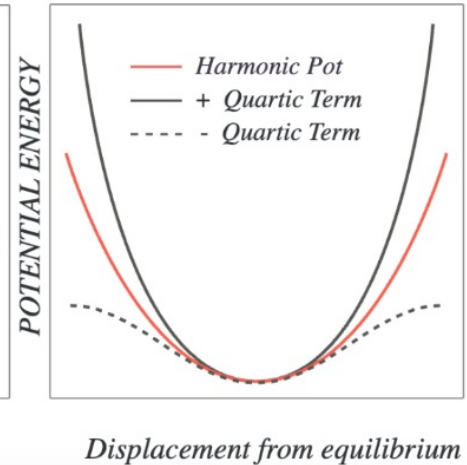
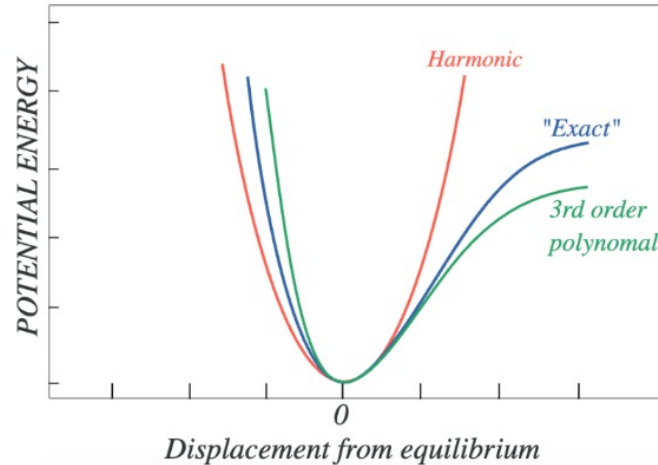
$$V(x) = \frac{1}{2}kr^2 + \frac{1}{6}\gamma r^3 \quad (5.2.5.2)$$

Anharmonicity cont.

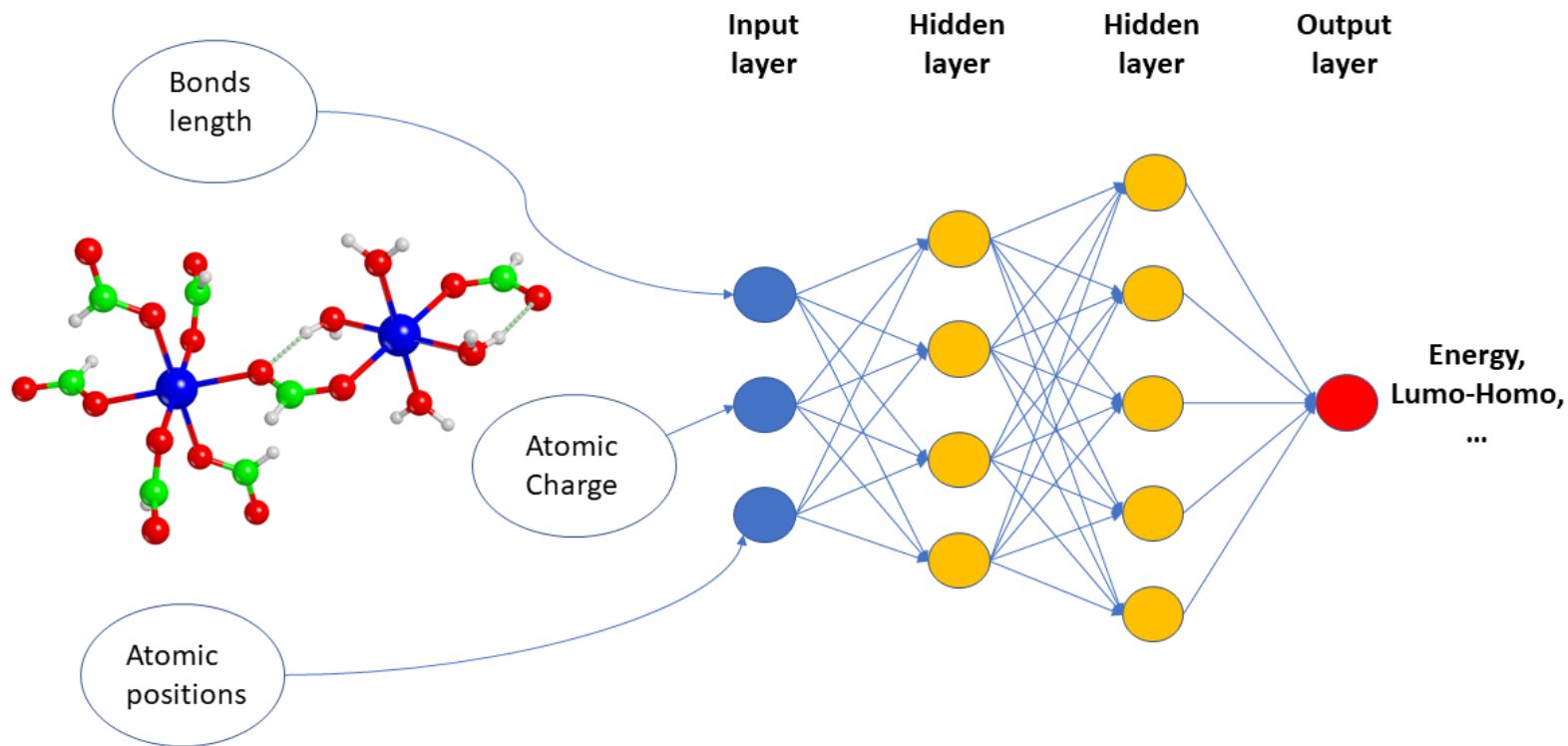
where

- $V(x_0) = 0$, and $r = R - R_0$.
- k is the harmonic force constant, and
- γ is the first (i.e., cubic) *anharmonic* term

It is important to note that this approximation is only good for R near R_0 .



Large systems into neural networks



Many-body interactions

Many different interactions can affect the energy at one “site” or atom

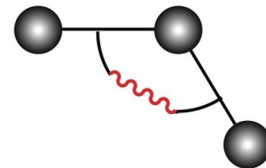
Direct neighbors have a large impact on local energy, but so can second and third neighbors

$$E_i = V^{(1)}(\mathbf{r}_i) + \frac{1}{2} \sum_j V^{(2)}(\mathbf{r}_i, \mathbf{r}_j) + \frac{1}{3!} \sum_{jk} V^{(3)}(\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k) + \dots$$

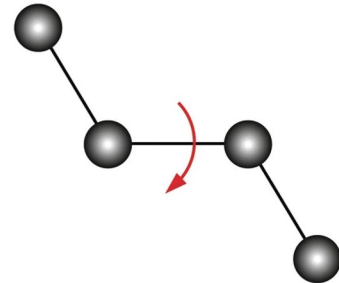
Bond stretching



Bond angle bending



Dihedral angle torsion



Van der Waal interaction



Electrostatic interaction

